On The Conservation Laws and Pathway Realizability of Biochemical Reaction Networks

Thanh-An Pham[1] and Tieu-Long Phan[1,2*]

[1]Bioinformatics Group, Department of Computer Science & Interdisciplinary Center for Bioinformatics & School for Embedded and Composite Artificial Intelligence (SECAI), Leipzig University, Härtelstraße 16–18, D-04107, Leipzig, Germany.
[2]Department of Mathematics and Computer Science, University of Southern Denmark, DK-5230, Odense M, Denmark.

*Corresponding author(s). E-mail(s): tieu@bioinf.uni-leipzig.de;
Contributing authors: thanh-an.pham@uni-leipzig.de;

Abstract

Background: Conserved metabolite pools and substrate availability constrain biochemical pathways before kinetic parameters are known. Analyzing these constraints across curated and rule-generated networks requires both reaction stoichiometry and the amounts of substrates consumed at each step. Species that appear on both sides of a reaction cancel from the net stoichiometry, although the reaction still needs them to fire.

Results: We present the chemical reaction network (CRN) submodule of `SynKit`, an open-source Python toolkit, for constructing biochemical reaction networks and analyzing their conserved pools, chemical reaction network theory (CRNT) properties, and realizable reaction sequences. `KEGG` pathways, supported `SBML` models, and rule-generated reactions share the same underlying representation. Twelve reference networks pass the validation checks and agree with `CRNT4SBML` on eight shared structural quantities. Ten reaction-bearing `BioModels` entries pass structural `SBML` round trips, including a model with 504 species and 827 directed reactions. Four `KEGG` modules illustrate conserved pools and the effects of model boundaries. In glycolysis, removing shared polyphosphate from a reaction changes deficiency and substrate requirements while preserving the net stoichiometric matrix. A 15-step glucose-to-pyruvate firing sequence is realizable after currency metabolites are removed but is blocked from glucose alone when they are retained. In a generated formose network, conservation coefficients match carbon counts for all 37 species, but equal carbon inventories differ in their ability to supply the selected tetrose and hexose routes. All reported structural enumerations complete within seconds, including all 60 configurations of the scaling study.

Conclusions: `SynKit` brings conservation analysis and reaction-by-reaction realizability checks to the same biochemical model. Its results help distinguish conserved quantities, unmet substrate requirements, and the assumptions needed for CRNT conclusions. Source code, recorded results, and reproducible analyses accompany the manuscript.

Keywords: Chemical reaction networks, chemical reaction network theory, Petri nets, conserved moieties, stoichiometry, pathway realizability, `SBML`

1 Background

Curated databases such as `KEGG` [1], `Reactome` [2], and `WikiPathways` [3] catalog known metabolic pathways and biochemical reactions, while graph-transformation methods enumerate candidate reactions from molecular seeds and transformation rules [4]. Before assigning kinetic parameters, structural analyses clarify the behavioral constraints imposed by network topology. Specifically, researchers examine which metabolite pools are conserved by stoichiometric invariants [5], which qualitative dynamical bounds or multistationarity criteria are governed by reaction network structure [6, 7], and whether a candidate pathway is realizable from a finite substrate inventory [8].

Assessing these constraints simultaneously is difficult because each analysis relies on a different mathematical abstraction of the biochemical system. A species graph captures connectivity between metabolites but loses the joint requirement of multiple co-substrates [9]. The net stoichiometric matrix records net changes in species abundance but omits participants that appear identically as reactants and products, such as catalysts or recycled cofactors. For example, a catalyzed reaction $A + C \rightarrow B + C$ and the uncatalyzed conversion $A \rightarrow B$ yield identical net columns in $\mathbf{S}$, yet only the first requires the catalyst C to proceed, and the two define different reaction complexes that alter network deficiency. Similarly, a balanced reaction combination or a qualitatively reachable product does not establish that the reactions can proceed in an admissible sequence from a finite inventory [8]. Representation choices during model import can therefore compromise the assumptions of another analysis.

Because of these distinct formulations, existing software packages specialize in separate subsets of analyses: `COBRApy` [10] supports constraint-based flux analysis [11, 12], `CRNT4SBML` [13] evaluates deficiency and potential bistability, `MØD` [4] explores rule-generated reaction spaces, `Tellurium` [14] and `libRoadRunner` [15] simulate kinetic ODEs, and Petri-net methods provide pathway realizability certificates [16, 17, 8]. Standard exchange formats such as `SBML` [18] and `COMBINE` [19], alongside converters like `KEGGtranslator` [20], facilitate model transfer. However, converting models between separate formats can inadvertently discard structural features, such as catalysts or boundary species, that specific analyses require.

Conserved pools and substrate sufficiency therefore need to be examined on the same underlying reaction model. In glycolysis [1], a route appears connected from glucose to pyruvate when cofactors are omitted, yet cannot proceed when ATP and redox cofactors are retained with finite initial pools. In a rule-generated formose network [21], every reaction may conserve carbon, yet a proposed pathway remains unrealizable if its reactions require more glycolaldehyde molecules than the initial precursor pool provides. In both cases, conservation laws alone cannot establish whether a reaction sequence is realizable from a specified finite inventory [8].

Here, we present the chemical reaction network (CRN) submodule of `SynKit` [22] to evaluate these constraints on a shared representation. By connecting `KEGG` pathway import, structural `SBML` exchange, and rule-based generation to exact conservation laws, deficiency theory, and Petri-net reachability, the toolkit tests whether reaction sequences can execute without discarding catalysts or cofactors. We validate the implementation across twelve reference networks against `CRNT4SBML`, verify structural round trips on ten `BioModels` entries, and benchmark exact enumeration limits. Finally, case studies in glycolysis and formose demonstrate how cofactor boundaries, participant retention, and precursor mixtures alter deficiency and pathway realizability even when net stoichiometry remains unchanged.

2 Implementation

2.1 Network representation and participant incidence

`SynCRN` serves as the core reaction-network data structure in `SynKit`, maintaining species, reactions, transformation rules, and chemical annotations (Fig. 1). It derives bipartite digraphs in `NetworkX`, stoichiometric matrices, and Petri nets for downstream structural analysis. The bipartite digraph has species vertices X and reaction vertices $\mathscr{R}$, and each reaction vertex is linked to its reactants and products. It is therefore the incidence representation of a directed hypergraph in which every reaction is a hyperedge, so multi-substrate requirements remain joint hyperedge tails rather than pairwise species edges.

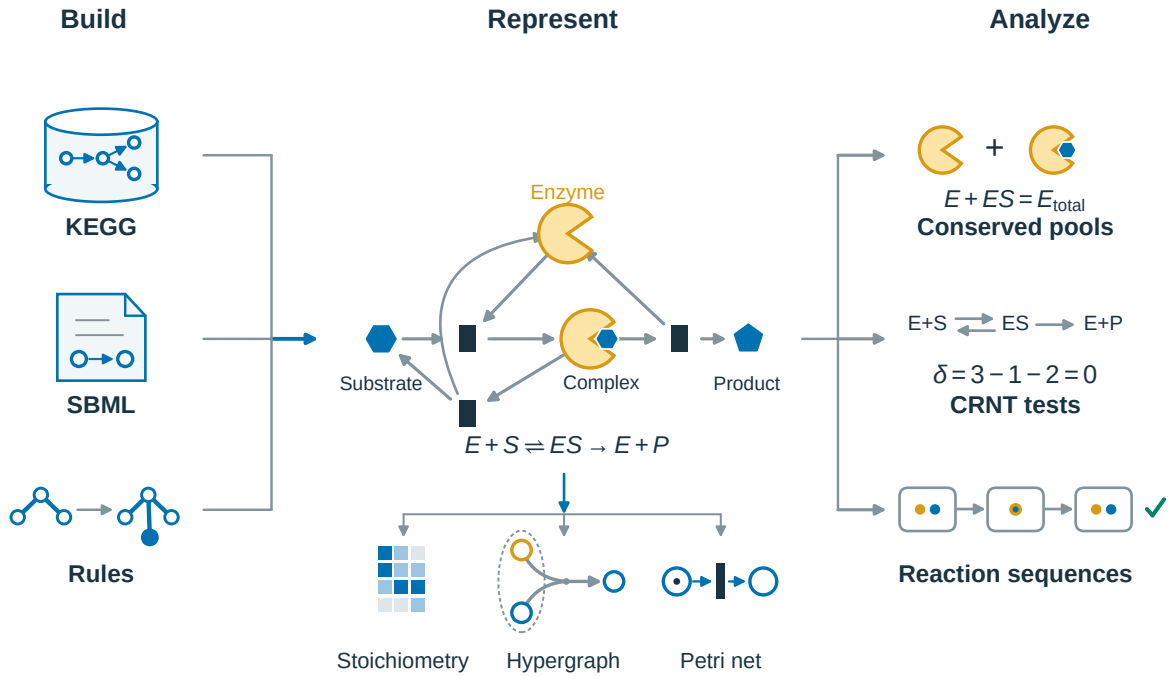


Fig. 1: Overview of `SynKit CRN` workflows across construction, representation, and structural analysis. **Build:** Reaction networks imported from curated databases (`KEGG`), exchange files (`SBML`), or generated by rule expansion are mapped to a unified data structure. **Represent:** An illustrative enzyme mechanism ($E+S \rightleftharpoons ES \rightarrow E+P$) is converted into three complementary views: stoichiometric matrices, directed hypergraphs, and Petri nets, ensuring that multi-reactant requirements (E and S) are captured as joint inputs. **Analyze:** These structural representations enable exact conservation analysis (identifying the conserved pool $E + ES$), chemical reaction network deficiency calculation ($\delta = 0$), and discrete pathway realizability through verified firing sequences. Blue circles denote metabolites, orange circles denote enzymes, and dark bars represent reactions.

For reaction r, reactant and product multisets have coefficients s_{xr}^{-} and s_{xr}^{+}, defining separate reactant and product (pre/post) incidence matrices $\mathbf{S}^{-} = (s_{xr}^{-})$ and $\mathbf{S}^{+} = (s_{xr}^{+})$ alongside the net stoichiometry $\mathbf{S} = \mathbf{S}^{+} - \mathbf{S}^{-}$. Retaining participants present on both sides in both matrices preserves catalytic and cofactor requirements. Their net coefficients cancel in $\mathbf{S}$, but they still govern reaction enabling and define distinct reaction complexes (Fig. 2a,b). In Petri-net terms, such shared species form self-loops, and the net incidence matrix alone cannot determine whether transitions are enabled [17]. Construction does not establish elemental balance automatically, so chemical interpretation requires suitable input equations and

annotations. The conditions under which a reaction network represents genuine chemical transformations are themselves a subject of study [23].

Reaction networks can be constructed from curated databases, generative chemistry, or standard model files. The `Query` module parses reaction equations from `KEGG`, `Construct` expands networks from chemical seeds using reaction rules, and `IO` imports models directly from `SBML`. During construction, reversible reactions can be partitioned into directed steps. Users can also choose whether to explicitly retain cofactors (such as ATP and NADH) and boundary species, which directly dictates the resulting conserved moieties, network deficiency, and discrete pathway realizability.

2.2 Conserved moiety and stoichiometric analysis

The stoichiometric matrix $\mathbf{S}$ records the net change in every species for each reaction. For concentrations c and reaction rates $v(c)$, the dynamics are written as $\dot{c} = \mathbf{S}v(c)$. A vector y satisfying $y^T\mathbf{S} = 0$ defines a conserved linear quantity $y^T c$, because the reactions can redistribute the participating species without changing this weighted total. `SynKit CRN` computes a rational basis for the left nullspace of $\mathbf{S}$ and converts its vectors to primitive integer representatives. These vectors describe all rational conservation relations, but they need not be non-negative or identify the smallest conserved pools.

Minimal P-semiflows are computed separately by exact Farkas elimination [24]. They are non-zero, non-negative integer vectors in the left kernel of $\mathbf{S}$ with inclusion-minimal supports, normalized by their greatest common divisor. Their supports identify minimal conserved pools of the represented network [5]. A P-semiflow guarantees mathematical conservation, whereas assigning its support to a chemical moiety requires species annotations and biochemical interpretation.

T-semiflows satisfy $\mathbf{S}x = 0$ with $x \geq 0$ and describe balanced combinations of reactions whose total net species change is zero. They are reaction-space relations, not conserved species pools (Fig. 2c). A T-semiflow fixes neither a firing order nor an initial marking, and it does not imply thermodynamic feasibility. All-positive left and right kernel vectors define conservativity and consistency, respectively. To determine positive-kernel feasibility, a linear program proposes a candidate vector that is accepted only after exact rational verification. Feasible networks are certified by an exact all-positive vector, infeasible networks by a rational separating vector, and unresolved solver outcomes remain undecided. Numerical kernel and Jacobian-pattern checks can also characterize approximate subspaces and qualitative matrix patterns, but because floating-point estimates do not isolate integer minimal supports, numerical rounding cannot substitute for exact conservation analysis.

2.3 Deficiency analysis and multistationarity bounds

Complexes are distinct reactant or product multisets. With n complexes, ℓ connected components of the complex graph (linkage classes), and stoichiometric rank s, deficiency is $\delta = n - \ell - s$. A network is weakly reversible when every linkage class is strongly connected. `SynKit CRN` calculates ranks exactly over $\mathbb{Q}$ and

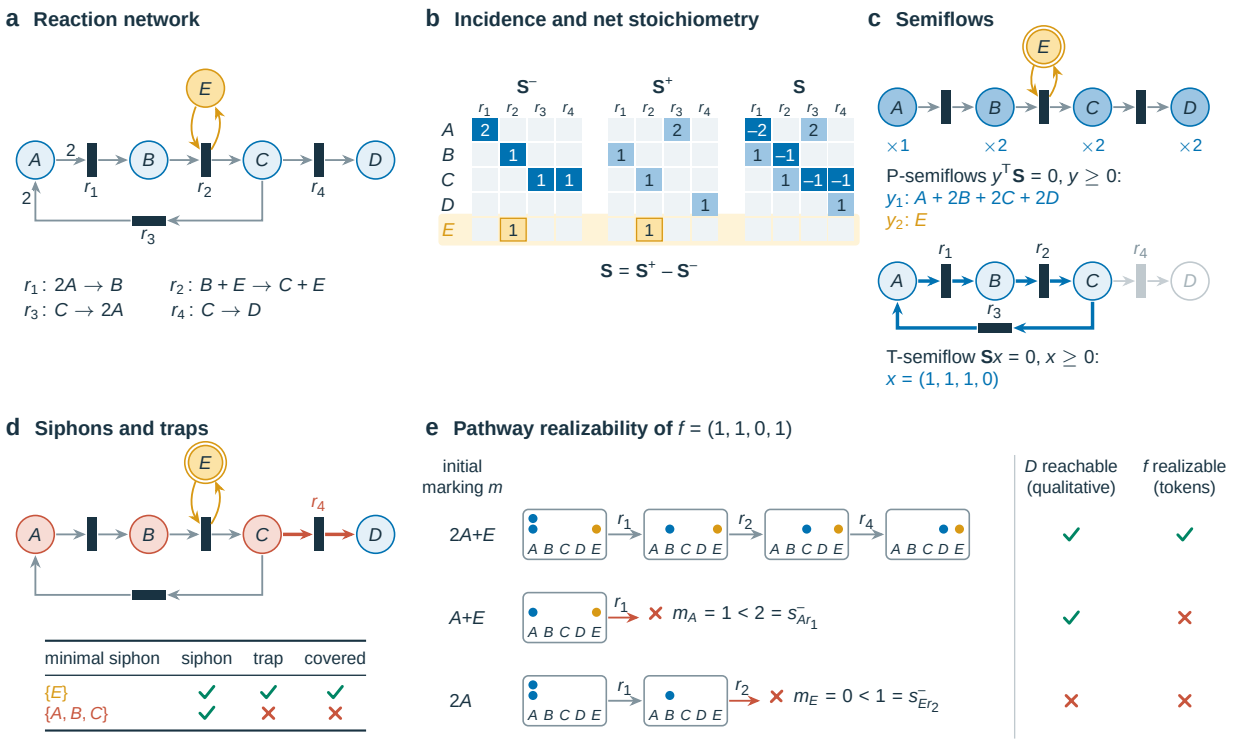


Fig. 2: Structural objects on one toy network, colored as in Fig. 1, with red marking blocked steps or uncovered sets. **(a)** Petri net (circles: species, bars: reactions, arc labels: coefficients other than one). The enzyme E is consumed and returned by r_2. **(b)** Reactant, product, and net incidence matrices. The row of E cancels in $\mathbf{S}$, so the net matrix alone does not show that r_2 requires E. **(c)** P-semiflows (shaded, with weights) are conserved pools. The T-semiflow (heavy arcs) is a balanced reaction combination that fixes no firing order. **(d)** Minimal siphons. $\{E\}$ contains a conserved pool and is covered, whereas $\{A, B, C\}$ drains through r_4 (heavy) and is uncovered. **(e)** Firing from three initial markings (dots: tokens). From $A + E$, qualitative reachability reaches D although no reaction is enabled. Inequalities give the failed condition $m_x \geq s^-_{xr}$.

reports the deficiency of the network and of each linkage class, weak reversibility, and the terminal strong linkage classes.

For positive mass-action rate constants, a weakly reversible deficiency-zero network has exactly one positive steady state in each positive stoichiometric compatibility class $(c_0 + \operatorname{im}\mathbf{S}) \cap \mathbb{R}^X_{>0}$, and it is locally asymptotically stable relative to that class [25, 6]. A deficiency-zero network that is not weakly reversible has no positive steady state. The Deficiency One Theorem gives at most one positive steady state per positive stoichiometric compatibility class when each linkage-class deficiency is at most one, their sum equals δ, and each linkage class has one terminal strong linkage class, while weak reversibility additionally ensures existence [26]. Failure of these hypotheses leaves the corresponding theorem undecided. Introductions to this body of theory for biologists are available [7]. Criteria that reach beyond deficiency, including injectivity [27], concordance [28], and graph-theoretic multistationarity tests [29], are not implemented here, and a network left undecided by the deficiency theorems is not thereby shown to be multistationary.

2.4 Pathway realizability and siphon analysis

A siphon is a non-empty species set for which every reaction producing a member also consumes a member. Once empty, it cannot refill under Petri-net firing semantics. A trap has the dual property that every reaction consuming a member also produces a member, so a marked trap cannot become empty [17]. `SynKit CRN` enumerates minimal siphons using implication propagation and closure minimization, within the branch-and-bound approach to structural enumeration [30]. Producing a selected species forces selection of a reactant. Selected and excluded species are propagated before branching. Each reported minimal set excludes its supersets from further search, and a maximal-closure deletion test verifies minimality. Traps use the dual incidence (Fig. 2d). This prunes many subsets but does not remove worst-case combinatorial growth.

The package checks whether every minimal siphon contains the support of a P-semiflow, and we call such a siphon covered. Under the boundedness and kinetic assumptions of the corresponding persistence result, this provides a sufficient persistence certificate for positive initial conditions [31]. An uncovered siphon means the test is insufficient, and does not prove extinction or identify a unique nutrient supplementation strategy. The returned details expose these uncovered sets.

A marking m specifies the non-negative integer inventory of each species. Reaction r is enabled precisely when $m_x \geq s_{xr}^-$ for every species, and firing gives $m' = m + \mathbf{S}_{\cdot r}$. Qualitative reachability instead accumulates reachable species sets and ignores consumption and multiplicity. Given a requested flux $f \in \mathbb{N}_0^{|\mathscr{R}|}$, realizability search seeks an enabled sequence firing each reaction exactly f_r times [8]. A positive result includes a certificate consisting of an ordered reaction list that can be checked by updating the inventory after every step. An exhaustive negative result is `unrealizable`, while reaching a state or depth limit is `inconclusive`. Fig. 2e contrasts the two notions on one requested flux. Certificates establish firing in the specified discrete model only.

2.5 Implementation and software comparison

The software employs a hybrid architecture where users interact entirely through a high-level Python interface, while computationally intensive tasks are transparently dispatched to an optimized C++17 library. These native routines compute exact rational nullspaces, CRNT deficiency invariants, minimal semiflows, siphon enumeration, and discrete firing sequences. To eliminate floating-point rounding errors and ensure exact certificates, linear-algebraic and combinatorial routines use arbitrary-precision integer arithmetic via Boost.Multiprecision. For generative chemistry, reaction spaces are expanded from seed molecules using `SMARTS` or double-pushout graph transformation rules via a native C++ process linked to `RDKit`. Following the `MØD` strategy framework [32], each rule application must use at least one newly generated molecule.

To ensure interoperability with the wider modeling community, `SynKit` provides native `SBML` Level 3 Core import and export. The parser preserves species identifiers, compartments, and exact stoichiometries across export and re-import, omitting kinetic rate laws and package extensions. Exported models provide

the structural foundation for parameterization in kinetic simulators like `Tellurium` or constraint-based packages like `COBRApy`.

Through this design, `SynKit` occupies a distinct niche in the systems biology software ecosystem (Table 1). Established packages typically specialize in continuous kinetic simulation (`Tellurium`), flux-balance analysis (`COBRApy`), or algebraic deficiency testing (`CRNT4SBML`). While `crnpy` evaluates exact semiflows and `MolFoundry` solves rule-based hyperflows, `SynKit` unifies rule generation, deficiency, minimal siphons, and discrete firing certificates on a shared model. Downstream analyses can then build directly on exported `SBML` networks. Section 1 of the Supplementary Information and Listing S1 provide complete implementation details and usage examples.

Table 1: Feature comparison of biochemical reaction-network software (✓ supported, (✓) partial or indirect as detailed in footnotes, × not supported). Capabilities are determined from official documentation and source code. `CRNT4SBML` is experimentally evaluated on shared benchmark networks (Table 2); formose generation is compared with `MØD` and `MolFoundry`, and fixed-route realizability with `MolFoundry` (Section 3.5). The `SynKit` column describes its chemical reaction network (CRN) submodule.

	`SynKit` (this work)	`CRNT4SBML` [13]	`CoNtRol` [33]	`crnpy` [34]	`COBRApy` [10]	`MØD` [4]	`MolFoundry` [35]	`Tellurium` [14, 15]
Network construction and exchange								
`KEGG` pathway import	✓	×	×	×	×	×	×	×
`SBML` import	(✓)[a]	✓	×	✓	✓	×	×	✓
`SBML` export	(✓)[a]	×	×	✓	✓	×	✓	✓
Rule-based network generation	✓	×	×	×	×	✓	✓	×
Structural analysis								
Deficiency and weak reversibility (CRNT)	✓	✓	✓	✓	×	×	×	×
Conservation laws (left-nullspace basis)	✓	✓	×	✓	×	×	×	✓
Minimal semi-positive conservation laws	✓	×	×	✓	×	×	×	×
Elementary flux modes or T-semiflows	✓	×	×	✓	×	×	×	×
Minimal siphons and traps	✓	×	(✓)[b]	×	×	×	×	×
Dynamics and pathway feasibility								
Multistationarity analysis	×	✓	✓	(✓)[c]	×	×	×	×
Pathway realizability (firing sequence)	✓	×	×	×	×	(✓)[d]	✓[f]	×
Constraint-based modeling (e.g., FBA)	×	×	×	×	✓	×	(✓)[f]	×
Kinetic simulation	×	(✓)[e]	×	×	×	×	✓[g]	✓

[a] `SBML` Level 3 Version 2 Core reaction structure only, with kinetic laws and rules not handled.
[b] Siphons underlie persistence tests, but minimal-siphon enumeration is not exposed as an output.
[c] `crnpy` supplies Gröbner-basis, elementary-mode, and ACR machinery rather than bistability search.
[d] `MØD` evaluates integer hyperflows rather than discrete firing certificates over a finite marking.
[e] ODE simulation is provided in support of `CRNT4SBML`'s bistability analysis.
[f] `MolFoundry` solves integer hyperflows over derivation graphs (Gurobi or SMT) for firing orders.
[g] Rule-based stochastic simulation with on-demand network expansion.

2.6 Validation and benchmark design

We validate the toolkit through analytical benchmarks, runtime scaling experiments, and biochemical case studies. First, structural correctness is evaluated across twelve reference networks, including enzyme mechanisms, multisite phosphorylation cycles [36], the bistable Edelstein switch [37], and the Shinar–Feinberg robustness motif [38], with shared CRNT invariants independently confirmed by `CRNT4SBML`. Second, computational scaling for exact siphon and conserved-pool enumeration is profiled across synthetic topologies to establish practical tractability limits. Third, structural `SBML` exchange and deficiency preservation are evaluated across eleven curated `BioModels` entries spanning up to 504 species and 827 directed reactions.

Finally, two mechanistic case studies demonstrate how structural representation choices dictate pathway feasibility from finite metabolite pools. In `KEGG` glycolysis, we assess how retaining catalytic polyphosphate and closing adenine/nicotinamide cofactor cycles govern discrete pathway execution. In rule-generated formose chemistry, we contrast qualitative hypergraph reachability with stoichiometric realizability under finite precursor budgets. Each feasible pathway is confirmed by an ordered reaction sequence from available precursors, while infeasibility is proved through exhaustive search.

3 Results and discussion

3.1 Reference-network validation and comparison with `CRNT4SBML`

All twelve reference networks match their analytical and algebraic ground-truth values (Table 2; sources and validation objectives detailed in Table S1). The benchmark suite spans classic enzyme kinetics (Michaelis–Menten and its reversible counterpart), multisite phosphorylation cycles [36], the bistable Edelstein switch [37], and the Shinar–Feinberg robustness motif [38]. In addition, constructed diagnostic cases verify boundary conditions: disconnected reversible pairs confirm that independent conserved pools are correctly isolated without conflating disjoint subnetworks, while the open reactor ($0 \to A \to B \to 0$) tests exact handling of inflow and outflow through the zero complex.

Table 2: Structural invariants of the twelve reference networks: species, directed reactions, complexes (n), linkage classes (ℓ), stoichiometric rank (s), deficiency (δ), conservation dimension (laws), and weak reversibility (WR). Every network matches its analytical and algebraic ground truth, and `CRNT4SBML` reproduces all eight shared structural quantities.

Network	Species	Reactions	n	ℓ	s	δ	Laws	WR
Reversible isomerization	2	2	2	1	1	0	1	yes
Irreversible isomerization	2	1	2	1	1	0	1	no
Isomerization cycle	3	3	3	1	2	0	1	yes
Disconnected reversible pairs	6	6	6	3	3	0	3	yes
Pairs with stoichiometry	6	6	6	3	3	0	3	yes
Michaelis–Menten	4	3	3	1	2	0	2	no
Reversible Michaelis–Menten	4	4	3	1	2	0	2	yes
Edelstein switch	3	6	5	2	2	1	1	yes
Phosphorylation: one site	6	6	6	2	3	1	3	no
Phosphorylation: two sites	9	12	10	2	6	2	3	no
Shinar–Feinberg motif	2	2	4	2	1	1	1	no
Open reactor	2	3	3	1	2	0	0	yes

To benchmark structural interoperability, `CRNT4SBML` 0.0.15 independently parsed all twelve exported `SBML` models. Across all twelve benchmarks, both tools yielded identical results for all eight shared structural quantities (Table 2), confirming topological consistency across independent implementations. Furthermore, standard floating-point nullspace methods produce numerical residuals on all closed networks, where naive scaling and rounding fail to recover valid integer conservation vectors on three of the twelve models (Table S2). In contrast, `SynKit`'s exact rational arithmetic guarantees that the conservation identity $y^T \mathbf{S} = 0$ holds strictly across all networks without numerical rounding error.

3.2 Computational cost of exact structural enumeration

Enumerating minimal siphons (Fig. 2d) and exact conserved pools (Fig. 2c) identifies extinction-prone species sets and mass conservation laws, but naive search scales exponentially with $2^N - 1$ candidate subsets. To prevent combinatorial explosion, implication pruning propagates stoichiometric constraints along active reactions during search (Fig. 3a). In reversible chains, selecting an initial species forces each adjacent neighbor, isolating the entire chain as the unique minimal siphon without evaluating spurious sub-combinations. Across five repeats per size, pruned search returns identical siphon sets with a 45.3-fold speedup at 21 species (0.79 ms versus 35.61 ms for exhaustive search, Fig. 3b).

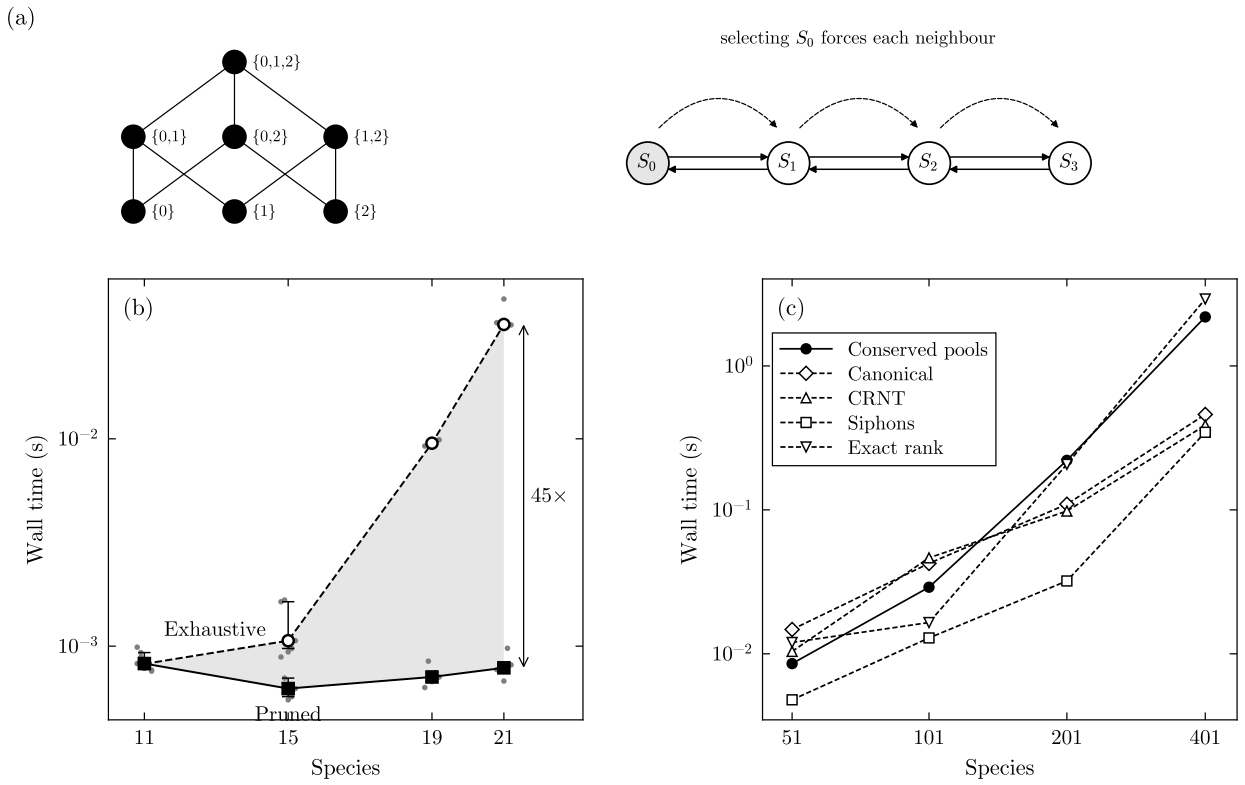


Fig. 3: Search algorithms and runtime scaling on reversible-chain networks. **(a)** Schematic of exhaustive power-set search versus implication pruning, where selecting S_0 forces all chain species into the minimal siphon. **(b)** Minimal-siphon runtimes across five repeats per size (points: individual runs, markers: medians and IQRs, bracket: speedup ratio at 21 species). **(c)** Wall times for five structural analyses across chains up to 401 species.

The broader benchmark suite evaluates scalability across directed chains, reversible chains, and random sparse bimolecular networks (Fig. S3). All 60 requested configurations complete without numerical failure. On the 401-species reversible chain, minimal siphons (Fig. 2d) take 0.346 s, canonicalization 0.46 s, exact conserved pools (Fig. 2c) 2.19 s, and standalone exact rank 2.92 s (Fig. 3c). The random networks with 49, 99, 197, and 395 species contain 331, 272, 47, and 87 minimal siphons, completing siphon and conserved-pool enumeration in 0.158 s and 0.0514 s at 395 species. Stoichiometric implication pruning and rational nullspace methods thus keep exact structural enumeration computationally practical on networks with hundreds of species.

3.3 Structural SBML round-trip validation in BioModels

We selected eleven BioModels [39] entries spanning a range of model sizes: ten reaction-bearing models and one SBML-qual negative control. Source identifiers, URLs, and input checksums accompany the results. All ten reaction-bearing models pass the supported structural round-trip checks, and the computed deficiency values are consistent with the complex and linkage-class counts. The largest model, BIOMD0000000255, contains 504 species and 827 directed reactions. Structural enumeration identifies 75 minimal siphons and 17 minimal conserved pools in 0.0426 s and 3.05 s, respectively (Fig. 4b). All ten models complete both enumerations, and every enumerated siphon is checked for coverage. Neither BIOMD0000000255 nor the 100-species BIOMD0000000019 (20 minimal siphons, 19 minimal conserved pools) satisfies the sufficient persistence condition: at least one minimal siphon is not covered by a conserved pool, so the condition does not certify persistence for these models. The SBML-qual control, BIOMD0000000562, is diagnosed as reaction-free, as expected for Core structural analysis. The set spans a range of sizes but does not estimate repository-wide support. Table S4 lists all eleven inputs and their structural results.

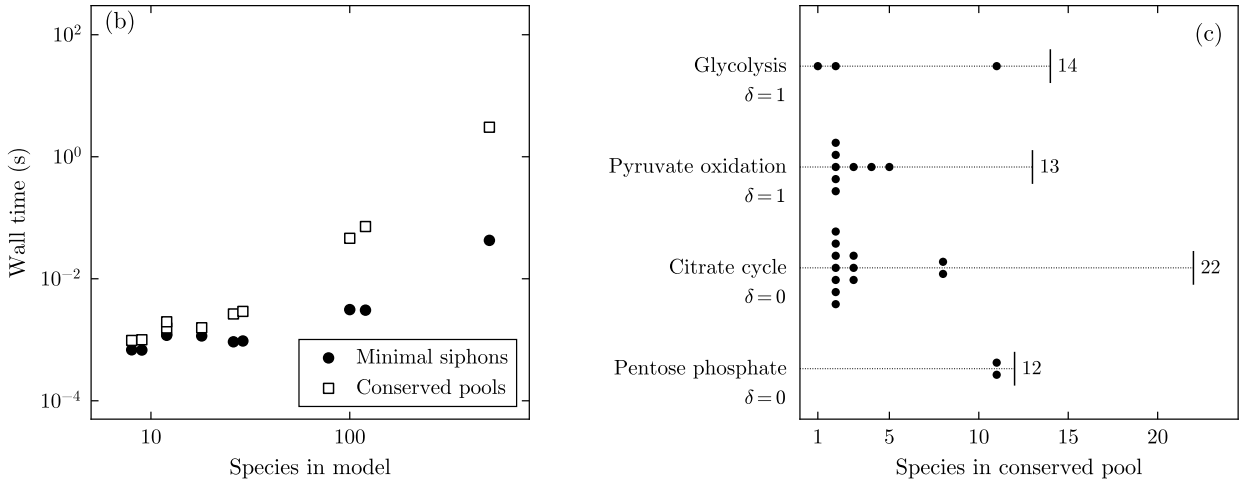


Fig. 4: Enumeration runtimes in BioModels and conserved pools in KEGG modules. **(b)** Wall time for completed minimal-siphon and exact conserved-pool enumeration in ten reaction-bearing BioModels models. All ten models complete both enumerations. **(c)** Minimal non-negative conserved pools in four reduced KEGG modules. Each dot is one pool placed at its number of species, pools of equal size are stacked, and the dotted line spans the module species, ending at the module size.

3.4 Conservation and cofactor requirements in glycolysis

Four cached KEGG [1] modules cover glycolysis (M00001), pyruvate oxidation (M00307), the citrate cycle (M00009), and the pentose phosphate pathway (M00004). Their conserved-pool supports appear in Fig. 4c. In the reduced citrate-cycle model, minimal non-negative conservation vectors recover nucleotide pairs, a quinone pair, a ferredoxin pair, coenzyme A, and a lipoyl pool. Because these interpretations reflect the retained reaction equations and labels, they do not imply organism-specific physiological pool conservation.

The reduced glycolysis model has 14 species and 15 directed reactions ($n = 16$, $\ell = 4$, $s = 11$, and $\delta = 1$). Its largest minimal conservation support spans eleven carbohydrate intermediates, assigning weight two to hexoses and one to trioses (Fig. 5). Glucose and oxidized ferredoxin form uncovered singleton siphons.

Polyphosphate forms a covered singleton because the cached generic-polymer reaction retains it on both sides. This symbolic representation does not resolve polymer chain length or prove catalytic phosphate chemistry.

For one glucose token, the prescribed flux has a 15-step firing certificate ending in two pyruvate tokens, and search explores 30 states. The repeated lower-pathway firings are specified in the requested flux and are compatible with the two trioses produced by aldolase. Certificate replay verifies enabling at each step and conservation of the weighted carbohydrate pool. Requesting only the lower pathway from glucose fails because the required intermediates are absent.

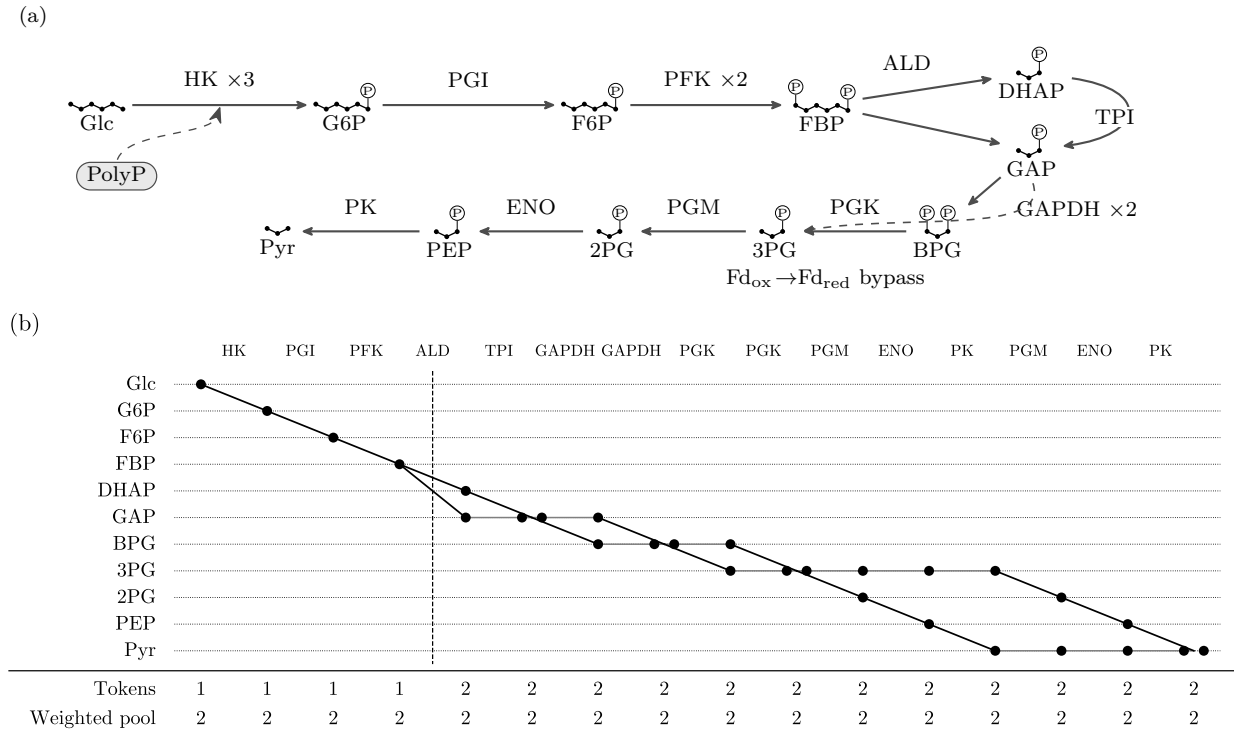


Fig. 5: Conserved pools and pathway realizability in reduced `KEGG` glycolysis (`M00001`). **(a)** Bipartite network (14 species, 15 directed reactions, currency metabolites omitted) retaining shared polyphosphate and the ferredoxin bypass, where edge labels indicate parallel alternatives. Each metabolite is drawn as its carbon skeleton (dots: carbon atoms, circled P: phosphate), so hexoses carry six carbons and trioses three. **(b)** Token markings across the certified 15-step conversion of one glucose to two pyruvates (each dot is one token, lines link each consumed intermediate to its products, column labels name the enzyme fired, and the dashed line marks aldolase cleavage). Bottom rows trace total carbohydrate tokens and the conserved weighted pool ($2 \times$ hexoses $+ 1 \times$ trioses $= 2$). Inactive polyphosphate and ferredoxin are omitted.

The controlled experiment isolates the information lost by cancellation (Fig. 6). Removing the common polyphosphate incidence from `R02189` preserves every species row, reaction column, and entry of $\mathbf{S}$. Consequently, its kernel and conservation space are unchanged. Yet the complex count changes from 16 to 14, linkage classes from four to three, and deficiency from one to zero. With one glucose and zero polyphosphate tokens, the retained reaction is blocked, whereas the cancelled reaction fires. Supplying one polyphosphate token enables the retained reaction. Two networks with the same net stoichiometric matrix can therefore differ in deficiency and in which reactions can fire.

A separate boundary control restores the currency metabolites, producing a 24-species model with the same 15 reactions. Under the same glucose-only inventory and requested flux, no reaction is enabled, because hexokinase lacks ATP, and exhaustive search returns `unrealizable`. The 15-step certificate therefore depends on currency removal. The reduced sequence establishes conversion of the carbohydrate intermediates, and evaluating cofactor sufficiency requires restoring the omitted species and specifying their initial amounts. Supplementary Section 5 gives the same glycolysis flux, firing certificate, and inventory controls in detail.

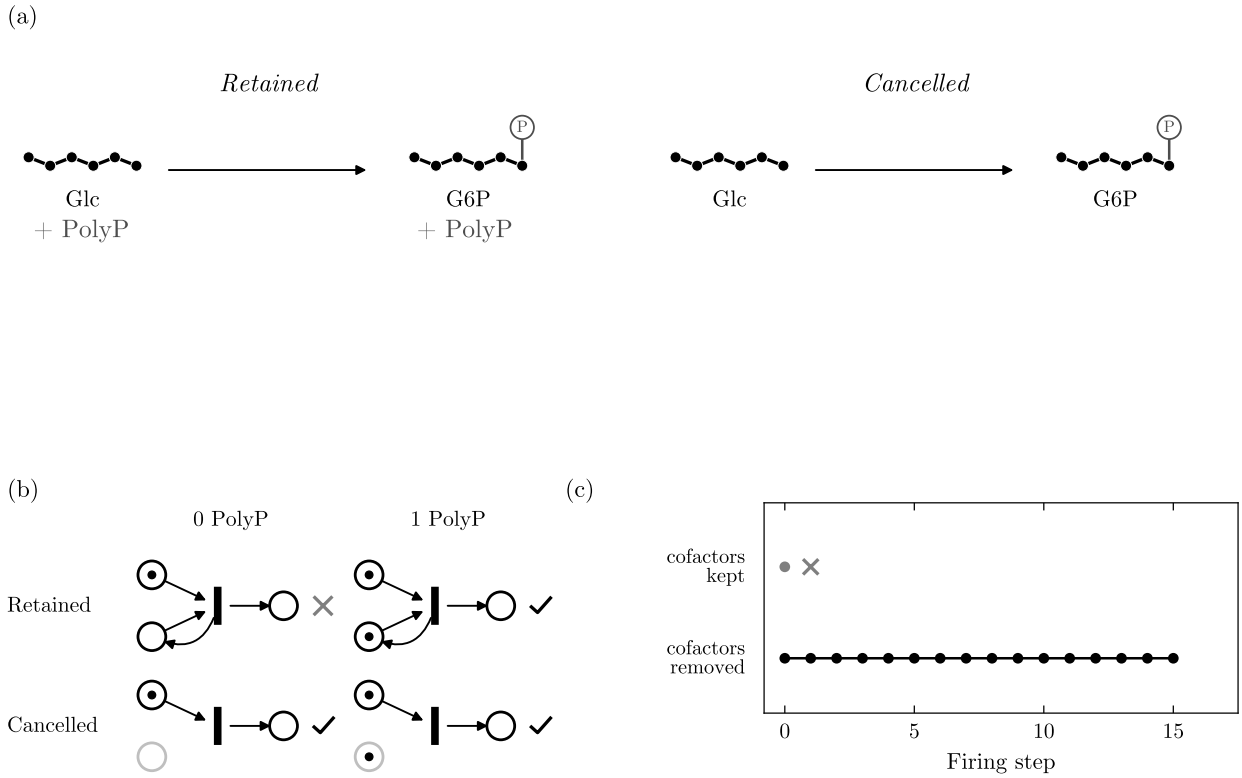


Fig. 6: Sensitivity of structural invariants and pathway realizability to participant retention and currency omission. **(a)** Cancelling shared polyphosphate (`PolyP`) in reaction `R02189` preserves net stoichiometry ($\mathbf{S}_{\text{retained}} = \mathbf{S}_{\text{cancelled}}$) but changes complex count ($n : 16 \to 14$), linkage classes ($\ell : 4 \to 3$), and deficiency ($\delta : 1 \to 0$). Glc indicates glucose, G6P indicates glucose 6-phosphate, filled circles represent carbon atoms, and P represents phosphate. **(b)** `R02189` as a Petri net under zero or one `PolyP` token (upper input place: glucose, lower input place: `PolyP`, output place: G6P, dots: tokens). The retained reaction consumes and returns `PolyP`, so it fires only when a `PolyP` token is present (check: enabled, cross: blocked). **(c)** Firing steps of the prescribed glycolytic flux from one glucose. With the currency metabolites (cofactors) kept in the model but not supplied (24 species), hexokinase lacks ATP, no reaction is enabled, and exhaustive search fails. With them removed (14 species), a certified 15-step sequence reaches two pyruvates.

3.5 Carbon conservation and precursor requirements in formose networks

The formose reaction, in which formaldehyde condenses to a mixture of sugars [21], gives a rule-generated counterpart to the curated `KEGG` modules. Starting from formaldehyde and glycolaldehyde, four rules for aldol addition, retro-aldol cleavage, and reversible keto-enol tautomerization generate 37 species and 44 reactions after four iterations. Median cumulative generation time is 0.00668 s across five warmed runs, and

the network reaches species containing ten carbon atoms. The iteration count is a rule-expansion depth, not physical reaction time. The network has 54 complexes, 17 linkage classes, rank 36, and deficiency one.

Matched expansions in `MØD` 1.0.0.75 [4] and `MolFoundry` 0.1.11 [35] independently validate network generation. At six iterations, all three tools yield identical sets of 10 572 species and 11 239 stoichiometric reactions. Median cumulative generation times are 4.29 s for `SynKit`'s native Lewis-hydrogen expander, 1.58 s for `MØD`, and 11.6 s for `MolFoundry`; Supplementary Table S10 gives per-depth timings, network-equivalence checks, and resource limits. These measurements include rule and seed parsing and network construction on one logical CPU. Supplementary Table S11 separates the Python/RDKit, native explicit-hydrogen, and native Lewis-hydrogen paths. No tool yields a complete depth-seven network under the stated limits.

The four-iteration network's one-dimensional left nullspace yields a positive primitive conservation vector. Comparing its coefficients with carbon counts from the molecular graphs gives exact agreement for all 37 species (Fig. 7). This is a consistency check between generated stoichiometry and molecular structures. Within this carbohydrate-only rule set, elemental counts can be proportional, but the result does not show that stoichiometry uniquely identifies elemental composition in general.

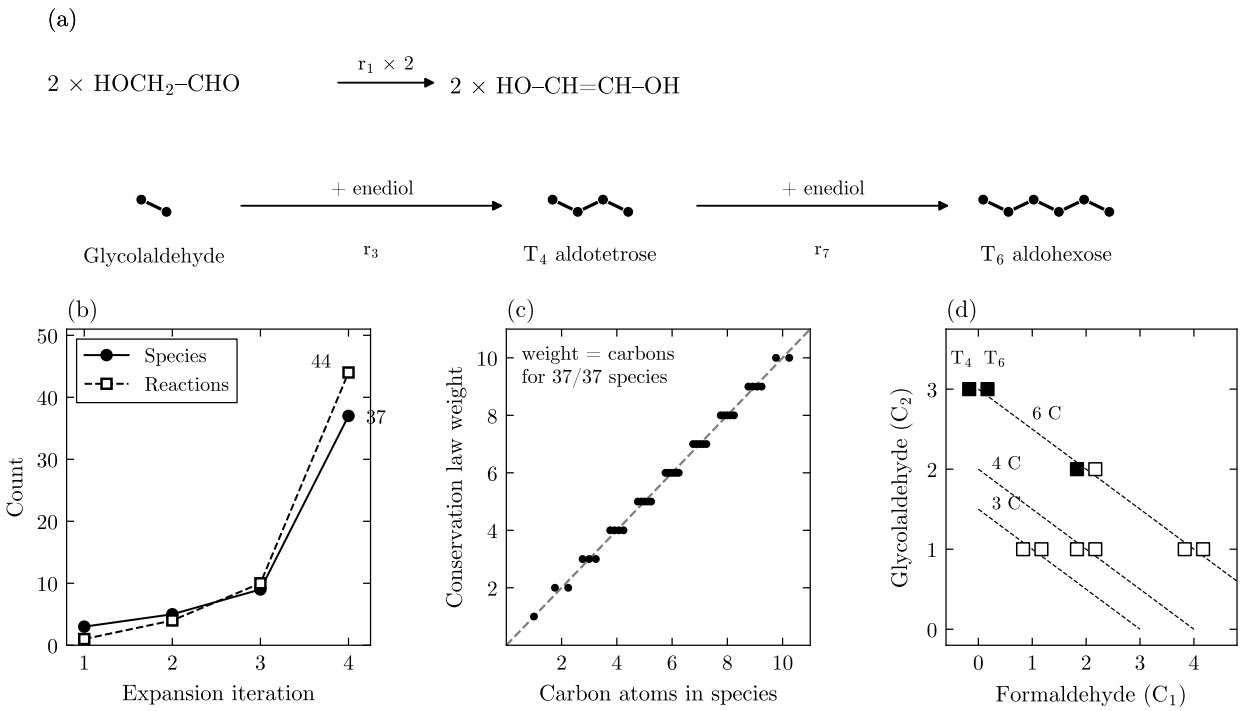


Fig. 7: Carbon conservation and precursor-dependent pathway realizability in the generated formose network. **(a)** Target hexose route requiring three glycolaldehyde precursors via two enolizations (r_1) and two aldol additions (r_3, r_7), each addition consuming one enediol (filled circles: carbon atoms). **(b)** Network growth across four rule-expansion iterations from formaldehyde and glycolaldehyde. **(c)** Weights of the single conservation law against carbon atoms per species. They agree for all 37 species, which confirms that the generated reactions conserve carbon (points jittered horizontally, dashed line: $y = x$). **(d)** Five initial inventories by formaldehyde and glycolaldehyde count, with dashed lines joining inventories of equal total carbon. The left and right squares of each inventory refer to the tetrose (T_4) and hexose (T_6) routes, filled when a certificate is verified and open when exhaustive search fails. Inventories with six carbon atoms differ in which routes they can supply.

Qualitative search reaches representative tetrose and hexose targets from the two seed species, but realizability distinguishes inventories that share the same total carbon (Fig. 7d). With four formaldehyde and one glycolaldehyde molecule, neither selected route is realizable. With two of each precursor, the tetrose route is realizable but the hexose route is not. With three glycolaldehyde molecules, both are. Each of these inventories contains six carbon atoms. The selected routes therefore require suitable precursor multiplicities as well as a sufficient carbon budget. `MolFoundry` independently agrees on all ten route–inventory decisions, and all three positive firing certificates pass independent replay. These results concern fixed reaction multisets in the generated network, and they do not establish unrestricted target unreachability or experimental synthetic feasibility. Supplementary Section 6 gives the route definitions and initial pools.

4 Conclusions

The CRN submodule of `SynKit` unites stoichiometric conservation analysis, chemical reaction network theory (CRNT), and Petri-net reachability on a shared reaction representation. Retaining explicit reactant and product incidences prevents the loss of catalysts and recycled cofactors that occurs under net stoichiometric reduction. Reference checks and `BioModels` evaluations demonstrate that exact deficiency, rank, and semiflow computations run reliably across curated, imported, and rule-generated reaction networks, while firing certificates verify execution from finite substrate inventories.

Several constraints define the scope of these analyses. Structural properties and reachability certificates evaluate topological enabling and finite-inventory realizability, but do not predict reaction rates, metabolite concentrations, thermodynamic driving forces, or in vivo activity. The conclusions depend on the reaction list, reversibility, boundary species, and cofactor treatment. Omitting cofactors removes their enabling requirements, whereas retaining them requires specifying their initial inventories and any exchange reactions. Exact enumeration of minimal siphons and semiflows can have combinatorial output and runtime growth, so resource limits and incomplete searches must remain explicit.

Future work should broaden input validation and improve scalable exact enumeration with explicit resource reporting. Coupling reachability certificates with estimated reaction free energies ($\Delta_r G$) will enable thermodynamic filtering of generated reaction spaces before kinetic modeling. Furthermore, integrating `SynKit`'s incidence representation with constraint-based flux methods and dynamic simulators will bridge topological invariants with quantitative kinetics, while native parallel expansion will scale exploration across complex metabolic and prebiotic chemistries.

Availability and requirements

Project name: `SynKit` (CRN submodule)

Project home page: github.com/TieuLongPhan/SynKit

Software version: `SynKit` 1.6.5 [40]

Operating system(s): Platform independent (Linux, macOS, Windows)

Programming language: Python, C++17, and C++20

Other requirements: Python $\geq$ 3.10, `NetworkX` $\geq$ 3.3, `NumPy`, `SciPy`, `RDKit`, Boost headers, CMake

License: MIT

Any restrictions to use by non-academics: None

List of abbreviations

ACR: Absolute concentration robustness

CRN: Chemical reaction network

CRNT: Chemical reaction network theory

DPO: Double pushout

EMP: Embden–Meyerhof–Parnas

FBA: Flux balance analysis

`KEGG`: Kyoto Encyclopedia of Genes and Genomes

`SBML`: Systems Biology Markup Language

`SMARTS`: SMILES arbitrary target specification

TCA: Tricarboxylic acid

WR: Weakly reversible

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Availability of data and materials

The CRN submodule is distributed with `SynKit` under the MIT License. Source code is available on GitHub at github.com/TieuLongPhan/SynKit [40], and the reproducible analysis scripts are located in the `Experiment/CRN` directory. `KEGG` [1] pathways remain subject to `KEGG` terms, and `BioModels` entries are distributed under the CC0 1.0 Public Domain Dedication.

Use of generative AI

Claude Opus (Anthropic) was used for language editing and TikZ figure formatting. The authors reviewed, verified, and take full responsibility for all manuscript content.

Competing interests

The authors declare that they have no competing interests.

Funding

This project has received funding from the European Union's Horizon Europe Doctoral Network programme under the Marie-Skłodowska-Curie grant agreement No 101072930 (TACsy, Training Alliance for Computational systems chemistry). Views and opinions expressed are however those of the author(s) only and do not necessarily reflect those of the European Union. Neither the European Union nor the granting authority can be held responsible for them.

Authors' contributions

Author contributions are reported using the CRediT taxonomy. **T.-A.P.**: Software, Validation, Writing (original draft), Writing (review and editing). **T.-L.P.**: Conceptualization, Methodology, Software, Validation, Writing (original draft), Writing (review and editing). Both authors read and approved the final manuscript.

Acknowledgements

Not applicable.

References

[1] Kanehisa M, Furumichi M, Sato Y, Matsuura Y, Ishiguro-Watanabe M (2025) KEGG: biological systems database as a model of the real world. Nucleic Acids Research 53(D1):D672–D677, DOI 10.1093/nar/gkae909

[2] Ragueneau E, Gong C, Sinquin P, Sevilla C, Beavers D, Grentner A, Griss J, Hogue GF, Li NT, Matthews L, et al (2026) The Reactome Knowledgebase 2026. Nucleic Acids Research 54(D1):D673–D681, DOI 10.1093/nar/gkaf1223

[3] Agrawal A, Balcı H, Hanspers K, Coort SL, Martens M, Slenter DN, Ehrhart F, Digles D, Waagmeester A, Wassink I, et al (2024) WikiPathways 2024: next generation pathway database. Nucleic Acids Research 52(D1):D679–D689, DOI 10.1093/nar/gkad960

[4] Andersen JL, Flamm C, Merkle D, Stadler PF (2016) A software package for chemically inspired graph transformation. In: International conference on graph transformation, Springer, Lecture Notes in Computer Science, vol 9761, pp 73–88, DOI 10.1007/978-3-319-40530-8_5

[5] Famili I, Palsson BO (2003) The convex basis of the left null space of the stoichiometric matrix leads to the definition of metabolically meaningful pools. Biophysical Journal 85(1):16–26, DOI 10.1016/S0006-3495(03)74450-6

[6] Feinberg M (1987) Chemical reaction network structure and the stability of complex isothermal reactors—i. the deficiency zero and deficiency one theorems. Chemical Engineering Science 42(10):2229–2268, DOI 10.1016/0009-2509(87)80099-4

[7] Gunawardena J (2003) Chemical reaction network theory for in-silico biologists. https://vcp.upf.edu/papers/crnt.pdf, preprint

[8] Andersen JL, Banke S, Fagerberg R, Flamm C, Merkle D, Stadler PF (2023) On the realisability of chemical pathways. In: Bioinformatics Research and Applications, International Symposium on Bioinformatics Research and Applications, vol 14248, Springer, pp 409–419, DOI 10.1007/978-981-99-7074-2_32

[9] Franzese N, Groce A, Murali T, Ritz A (2019) Hypergraph-based connectivity measures for signaling pathway topologies. PLoS Computational Biology 15(10):e1007,384, DOI 10.1371/journal.pcbi.1007384

[10] Ebrahim A, Lerman JA, Palsson BO, Hyduke DR (2013) COBRApy: COnstraints-Based reconstruction and analysis for Python. BMC Systems Biology 7:74, DOI 10.1186/1752-0509-7-74

[11] Orth JD, Thiele I, Palsson BØ (2010) What is flux balance analysis? Nature Biotechnology 28(3):245–248, DOI 10.1038/nbt.1614

[12] Schuster S, Hilgetag C (1994) On elementary flux modes in biochemical reaction systems at steady state. Journal of Biological Systems 2(02):165–182, DOI 10.1142/S0218339094000131

[13] Reyes BC, Otero-Muras I, Shuen MT, Tartakovsky AM, Petyuk VA (2020) CRNT4SBML: a Python package for the detection of bistability in biochemical reaction networks. Bioinformatics 36(12):3922–3924, DOI 10.1093/bioinformatics/btaa241

[14] Choi K, Medley JK, König M, Stocking K, Smith L, Gu S, Sauro HM (2018) Tellurium: An extensible Python-based modeling environment for systems and synthetic biology. BioSystems 171:74–79, DOI 10.1016/j.biosystems.2018.07.006

[15] Welsh C, Xu J, Smith L, König M, Choi K, Sauro HM (2023) libRoadRunner 2.0: a high performance SBML simulation and analysis library. Bioinformatics 39(1):btac770, DOI 10.1093/bioinformatics/btac770

[16] Zevedei-Oancea I, Schuster S (2003) Topological analysis of metabolic networks based on Petri net theory. In Silico Biology 3(3):323–345

[17] Murata T (1989) Petri nets: Properties, analysis and applications. Proceedings of the IEEE 77(4):541–580, DOI 10.1109/5.24143

[18] Hucka M, Finney A, Sauro HM, Bolouri H, Doyle JC, Kitano H, Arkin AP, Bornstein BJ, Bray D, Cornish-Bowden A, et al (2003) The systems biology markup language (SBML): a medium for representation and exchange of biochemical network models. Bioinformatics 19(4):524–531, DOI 10.1093/bioinformatics/btg015

[19] Bergmann FT, Adams R, Moodie S, Cooper J, Glont M, Golebiewski M, Hucka M, Laibe C, Miller AK, Nickerson DP, et al (2014) COMBINE archive and OMEX format: one file to share all information

to reproduce a modeling project. BMC Bioinformatics 15(1):369, DOI 10.1186/s12859-014-0369-z

[20] Wrzodek C, Dräger A, Zell A (2011) KEGGtranslator: visualizing and converting the KEGG PATHWAY database to various formats. Bioinformatics 27(16):2314–2315, DOI 10.1093/bioinformatics/btr377

[21] Butlerow A (1861) Bildung einer zuckerartigen Substanz durch Synthese. Justus Liebigs Annalen der Chemie 120(3):295–298, DOI 10.1002/jlac.18611200308

[22] Phan TL, González Laffitte ME, Weinbauer K, Merkle D, Andersen JL, Fagerberg R, Gatter T, Stadler PF (2025) SynKit: A graph-based Python framework for rule-based reaction modeling and analysis. Journal of Chemical Information and Modeling 65(24):13,012–13,019, DOI 10.1021/acs.jcim.5c02123

[23] Müller S, Flamm C, Stadler PF (2022) What makes a reaction network "chemical"? Journal of Cheminformatics 14(1):63, DOI 10.1186/s13321-022-00621-8

[24] Colom JM, Silva M (1991) Convex geometry and semiflows in P/T nets: A comparative study of algorithms for computation of minimal p-semiflows. In: Advances in Petri Nets 1990, Lecture Notes in Computer Science, vol 483, Springer, pp 79–112, DOI 10.1007/3-540-53863-1_22

[25] Horn F, Jackson R (1972) General mass action kinetics. Archive for rational mechanics and analysis 47(2):81–116, DOI 10.1007/BF00251225

[26] Feinberg M (1995) The existence and uniqueness of steady states for a class of chemical reaction networks. Archive for Rational Mechanics and Analysis 132(4):311–370, DOI 10.1007/BF00375614

[27] Craciun G, Feinberg M (2005) Multiple equilibria in complex chemical reaction networks: I. the injectivity property. SIAM Journal on Applied Mathematics 65(5):1526–1546, DOI 10.1137/S0036139904440278

[28] Shinar G, Feinberg M (2012) Concordant chemical reaction networks. Mathematical biosciences 240(2):92–113, DOI 10.1016/j.mbs.2012.05.004

[29] Mincheva M, Roussel MR (2007) Graph-theoretic methods for the analysis of chemical and biochemical networks. i. multistability and oscillations in ordinary differential equation models. Journal of Mathematical Biology 55(1):61–86, DOI 10.1007/s00285-007-0099-1

[30] Cordone R, Ferrarini L, Piroddi L (2005) Enumeration algorithms for minimal siphons in Petri nets based on place constraints. IEEE Transactions on Systems, Man, and Cybernetics – Part A: Systems and Humans 35(6):844–854, DOI 10.1109/TSMCA.2005.853504

[31] Angeli D, De Leenheer P, Sontag ED (2007) A Petri net approach to the study of persistence in chemical reaction networks. Mathematical Biosciences 210(2):598–618, DOI 10.1016/j.mbs.2007.07.003

[32] Andersen JL, Flamm C, Merkle D, Stadler PF (2014) Generic strategies for chemical space exploration. International Journal of Computational Biology and Drug Design 7(2/3):225–258, DOI 10.1504/IJCBDD.2014.061649

[33] Donnell P, Banaji M, Marginean A, Pantea C (2014) CoNtRol: an open source framework for the analysis of chemical reaction networks. Bioinformatics 30(11):1633–1634, DOI 10.1093/bioinformatics/

btu063

[34] Tonello E (2016) crnpy: A Python library for the analysis of chemical reaction networks. https://github.com/etonello/crnpy, accessed 7 November 2026

[35] Friedrichs M (2026) MolFoundry: rule-based graph transformation for chemistry, version 0.1.11. https://py.molfoundry.de/docs/, software documentation, accessed 7 November 2026

[36] Wang L, Sontag ED (2008) On the number of steady states in a multiple futile cycle. Journal of Mathematical Biology 57(1):29–52, DOI 10.1007/s00285-007-0145-z

[37] Edelstein BB (1970) Biochemical model with multiple steady states and hysteresis. Journal of Theoretical Biology 29(1):57–62, DOI 10.1016/0022-5193(70)90118-9

[38] Shinar G, Feinberg M (2010) Structural sources of robustness in biochemical reaction networks. Science 327(5971):1389–1391, DOI 10.1126/science.1183372

[39] Malik-Sheriff RS, Glont M, Nguyen TVN, Tiwari K, Roberts MG, Xavier A, Vu MT, Men J, Maire M, Kananathan S, Fairbanks EL, Meyer JP, Arankalle C, Varusai TM, Knight-Schrijver V, Li L, Dueñas-Roca C, Dass G, Keating SM, Park YM, Buso N, Rodriguez N, Hucka M, Hermjakob H (2020) BioModels—15 years of sharing computational models in life science. Nucleic Acids Research 48(D1):D407–D415, DOI 10.1093/nar/gkz1055

[40] Phan TL, Weinbauer K (2026) SynKit, version 1.6.5. https://github.com/TieuLongPhan/SynKit/releases/tag/v1.6.5, released 30 October 2026